

Practical 15 - Multinomial and Ordinal Logistic Regression

Statistical Models for Discrete Outcomes

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Datasets required: `multinomial.csv`

Introduction

In this practical we consider data from the 2003-2004 US National Health and Nutrition Examinations Survey (NHANES), fitting models that relate a categorised version of systolic blood pressure (SBP) to age and sex. We will categorise blood pressure according to SBP as follows.

Category	SBP
1. Normal	< 120
2. Prehypertension	$120 - 139$
3. High (stage 1)	$140 - 159$
4. High (stage 2)	≥ 160

Given the above classification system, it is of interest to model how the probability of falling into the four categories depends on age and sex.

Data

The dataset is called `multinomial.csv` and contains three variables as below.

Variable	Description
<code>sex</code>	“Female”, “Male”
<code>ageyrs</code>	Age in (whole number of) years

Variable	Description
sbp	Systolic blood pressure (mmHg)

Aims

- To understand how to interpret coefficients from multinomial and ordinal logistic regression models.
- To understand how to calculate estimated probabilities from multinomial and ordinal logistic regression models.
- To understand how to compare the fits of multinomial and ordinal logistic regression models.

Analysis

Question 1

Load the data into R. Then create a new categorical SBP variable **sbpcat**, categorised as above. One way to do this in R is as follows.

```
multi$sbpcat <- cut(multi$sbp,
                    breaks = c(0, 120, 140, 160, 300),
                    labels = c("Normal", "Prehypertension", "High1", "High2"),
                    right = FALSE,
                    ordered = T) # make it ordered
```

Tabulate **sbpcat** to see its distribution.

Also, to aid interpretation of the intercept estimates in our models, centre the age variable at 50 years.

```
multi$cen_age <- multi$ageyrs - 50
```

Question 2

Use the **multinom** function in the **nnet** package to fit a multinomial logistic model to the categorised SBP variable with sex and centred age as covariates.

Carefully describe the interpretation of each of the estimated parameters.

Question 3

From your fitted model, calculate (by hand) the estimated probability of having SBP in the second highest category (`spbcat = 3`) for a female aged 70 years.

Question 4

Use the `fitted.values` stored in your fitted model to plot the probabilities against age, first in females. Then create the same plot but for the males.

Check that the predicted probability of a 70 year old female having SBP in the second highest category (`spbcat = 3`) matches that calculated above.

Question 5

Now use the `polr` function in the `MASS` package to fit an ordinal logistic regression model to the categorised SBP variable with sex and centred age as covariates.

Carefully describe the interpretation of each of the estimated parameters.

Question 6

From your fitted ordinal logistic regression model, calculate (by hand) the estimated probability of having SBP in the second highest category (`spbcat = 3`) for a female aged 70 years.

Question 7

As for the multinomial model, use the `fitted.values` in your fitted ordinal logistic model object to plot the probabilities of being in each of the four SBP categories against age in females. Then repeat the plot but for the males.

Again check that the predicted probability of a 70 year old female having SBP in the second highest category (`spbcat = 3`) matches that calculated above.

Are the predictions from the ordinal and multinomial logistic regression models markedly different?

Question 8

To further assess whether the proportional odds assumption is reasonable here compare the log-likelihoods and AIC values from the two models.

Question 9

As a final check of the ordinal logistic regression model fit, compare the coefficients to those from a series of logistic regression models with different cut-offs. This can be done in different way. One is to create three new variables in the data frame for the three different dichotomisations:

1. Normal vs All other categories
2. Normal or Prehypertension vs Hypertension (Stage 1 or 2)
3. All other categories vs Stage 2 Hypertension

Alternatively, you can use the original continuous **sbp** variable together with the `I()` function to specify the different dichotomised outcomes directly in a call to `glm`.

Fit separate standard logistic regression models for each of the three dichotomised outcomes, with sex and centred age as covariates. Compare the coefficients from each model to the equivalent from the ordinal logistic regression model.

Considering all of the above, which model do you prefer for these data?